



Partnership for Safe Drinking Water Distribution Program Disinfectant Residual Update

September, 2025

The Bureau of Laboratory Services is responsible for administering compliance with the water quality monitoring requirements under both the federal Safe Drinking Water Act (SDWA) and state regulations. As monitoring locations that were temporarily not visited after the onset of COVID-19 are now being revisited, over 600 drinking water samples are collected from 74 monitoring locations.

Regarding disinfectant residual, the Partnership for Safe Water has a numerical goal that 95% of monthly routine samples be within the following range:

Total Chlorine Residual ≥ 0.50 mg/L and ≤ 4.0 mg/L

In the month of September 2025 6.8% of samples were less than 0.5 mg/L and therefore PWD failed to meet the 95% goal for Partnership. Prior to September 2025, PWD had not exceeded the Partnership goal since September 2016 (7.27%). Elevated temperatures may impact disinfectant residual in PWD's distribution system. The highest Partnership percentage in the previous three calendar years (2022 - 2025) was 6.8% in September 2025.

A total of 662 samples were collected in September 2025, down from 681 samples collected in August. 467 field chlorine residual samples were collected in accordance with DRR requirements. 465 field chlorine residual samples were collected in accordance with RTCR requirements. PWD adjusted the sampling schedule and frequency starting in the second half of March 2020 due to the ongoing COVID-19 pandemic. PWD is still collecting more than the required 360 samples for RTCR/DRR compliance. Starting in November 2020, PWD began to resume sampling from some routine fire and police stations for the RTCR/DRR compliance program. The selected fire and police stations are the locations that are sampled monthly as part of the DBP monitoring program as well. PWD also began resuming sample collection samples from additional fire and police stations earlier this year that were not being sampled during the first year of the pandemic. Starting in July 2020, PWD resumed monthly distribution Disinfection By-Product sampling from the normal fire and police station locations. As of October 2021, Chestnut Hill Hospital is no longer be sampled from and the Chestnut Hill Pressure District Pump Station replaced the hospital as the routine sampling location for the Chestnut Hill pressure district.

September 2025 had 45 samples below 0.50 mg/L throughout the distribution system, and 115 samples below 1 mg/l.

	Count	Description
Total DRR Samples for September 2025	467	
Number of Samples < 1.0 mg/L	115	Below PWD Distribution System goal of 1.00 mg/L
Number of Samples > 0.5 mg/L and < 1.00 mg/L	70	Meets PfSW goal of 0.50 mg/L. Does not meet PWD distribution system goal of 1.00 mg/L
Number of Samples > 0.15 mg/L and < 0.50 mg/L	44	Meets PADEP minimum distribution residual required (0.15 mg/L). Does not meet: PfSW goal (0.50 mg/L), PWD Distribution goal (1.00 mg/L)
Number of Samples < 0.15 mg/L	1	Does not meet: PADEP minimum distribution residual required (0.15 mg/L), PfSW goal (0.50 mg/L), PWD Distribution goal (1.00 mg/L)

Total THM values decreased compared to last month.0 locations had single sample TTHM results above the MCL of 80 ppb. Single sample TTHM results above the MCL are not an MCL exceedance as compliance with the MCL is based on Locational Running Averages.

HAA5 values decreased compared to last month.0 locations had single sample HAA5 results above the MCL of 60 ppb. Single sample HAA5 results above the MCL are not an MCL exceedance as compliance with the MCL is based on Locational Running Averages.

Below is a monthly range of DBP distribution values for the last 13 months. Values that exceed the MCL for a single sample are highlighted in red, and samples that exceed the operational goal of 75

MONTH	Minimum TTHM (µg/L)	Maximum TTHM (µg/L)	Minimum HAA5 (µg/L)	Maximum HAA5 (µg/L)
2025-Sep	38	61.9	17.5	26.7
2025-Aug	58.5	82.7	32.2	49.2
2025-Jul	60.3	90.3	2.14	64.5
2025-Jun	35.6	51.2	24	48.8
2025-May	36.4	53.1	22.6	46.4
2025-Apr	28.8	65.4	46.4	85.9
2025-Mar	26.4	37.7	37.8	56.5
2025-Feb	17.9	22.2	23.2	37.8
2025-Jan	18.3	28.7	25.5	41.6
2024-Dec	19.1	26.6	20.5	30.5
2024-Nov	25	51.9	18.6	29.6
2024-Oct	27.8	56.8	19.6	31.1
2024-Sep	40.1	60	24.5	37.1

TTHM LRAA MCL = 80 µg/L ; Operational Goal = 60 µg/L

HAA5 LRAA MCL = 60 µg/L ; Operational Goal = 45 µg/L

Percent of Grab Samples < 0.50 mg/L TCL2 Residual - Entire PWD Distribution System - by Month

Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
2025	0	0	0	0	0	0	1.7	2.2	6.8	NA	NA	NA
2024	0	0	0	0	0	0.5	0.5	1.3	1.3	0.4	0.2	0
2023	0	0	0	0	0	0	0.31	0.91	2.62	0.88	0	0
2022	0	0	0	0	0	0	0.15	1.06	0.87	0.15	0	0
2021	0	0	0	0	0	0	1.82	0	0	0.47	0	0
2020	0	0	0	0	0	0	0.57	1.86	2.78	0.65	0	0
2019	0	0	0	0	0.13	0.28	0.51	1.17	2.71	0.92	0	0

Key
>5.0 % Fails Partnership Goal
4.0 - 4.99 %
3.0 - 3.99 %
0.0 - 2.99 %